

AI ASSISTED CODING

ASSIGNMENT-7.3

2403A53002

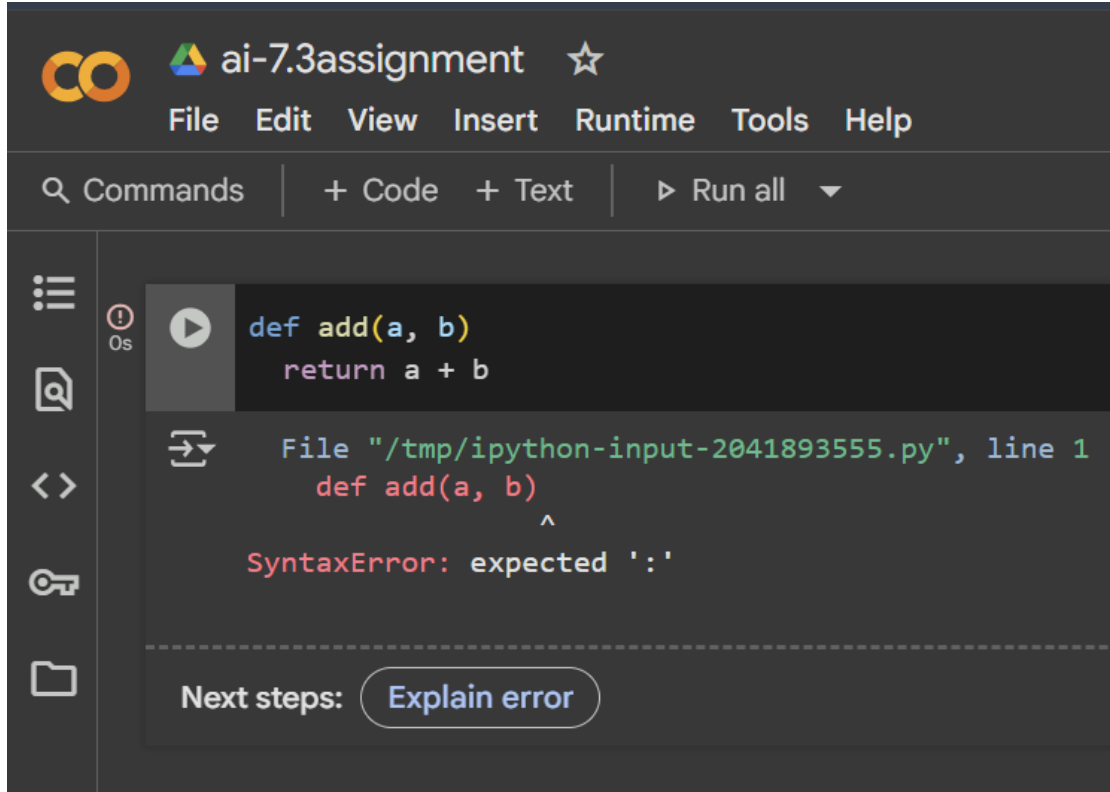
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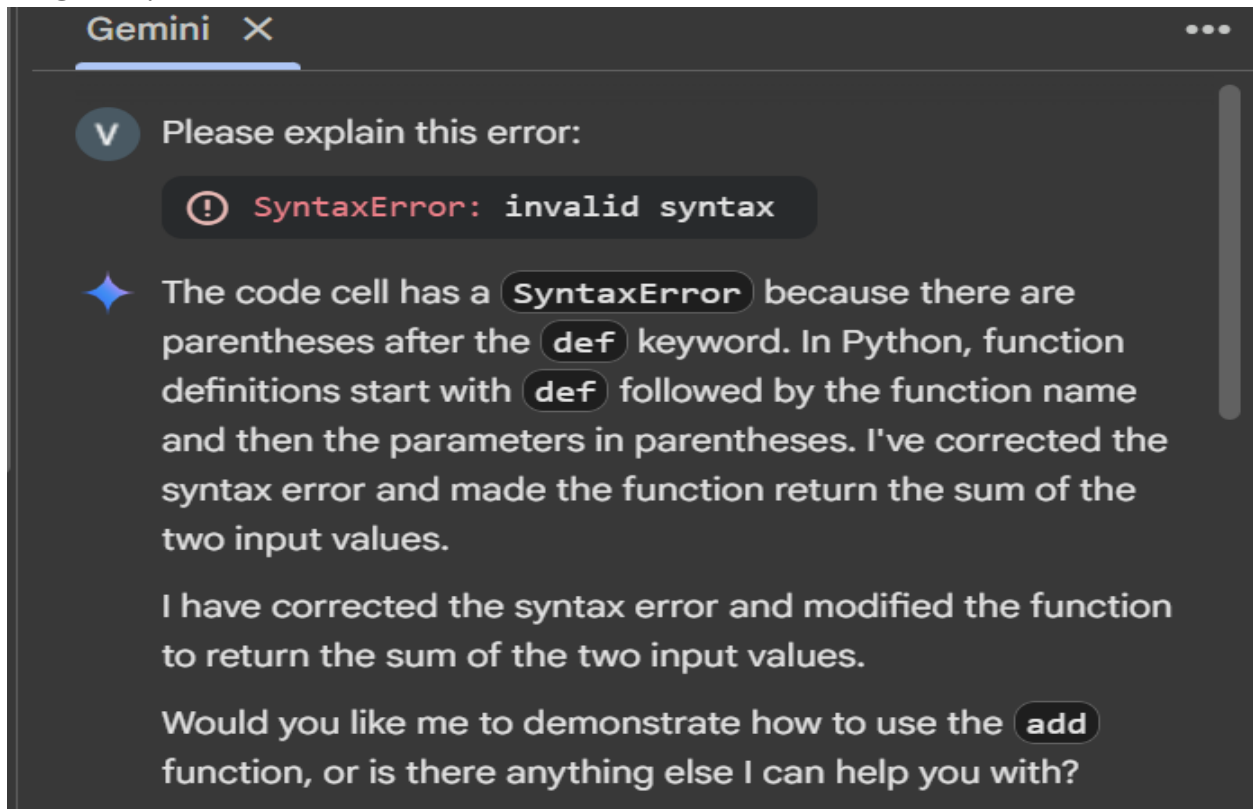
Task Description#1

- Paste a function with a missing colon (add(a, b)), and let AI fix the syntax error.

CODE:



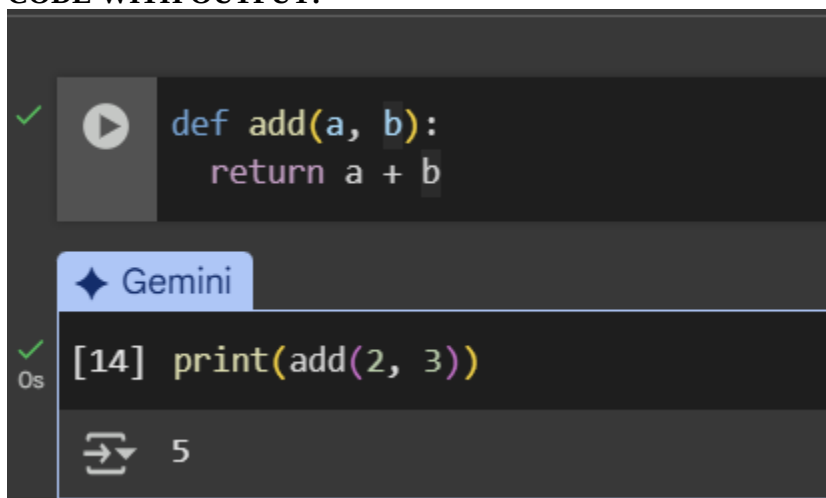
PROMPT:



CODE AFTER GENERATION:



CODE WITH OUTPUT:



Task Description#2 (Loops)

- Identify and fix a logic error in a loop that causes infinite iteration.

CODE:

```
def count_down(n):  
    while n>=0:  
        print(n)  
        n+=1
```

PROMPT:

• Identify and fix a logic error in a loop that causes infinite iteration.

✦ I can help with that! The current loop is infinite because the value of `n` is always increasing. To fix this, I will change the increment to a decrement so that the loop eventually terminates.

I have fixed the infinite loop issue by changing `n += 1` to `n -= 1`. Would you like me to run the corrected code?

👍 🗨

OUTPUT:

```
[13] def count_down(n):  
      while n>=0:  
          print(n)  
          n-=1  
  
      count_down(5)
```

◆ Gemini

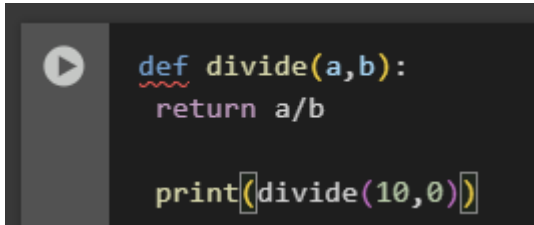
count_down(5)

5
4
3
2
1
0

Task Description#3

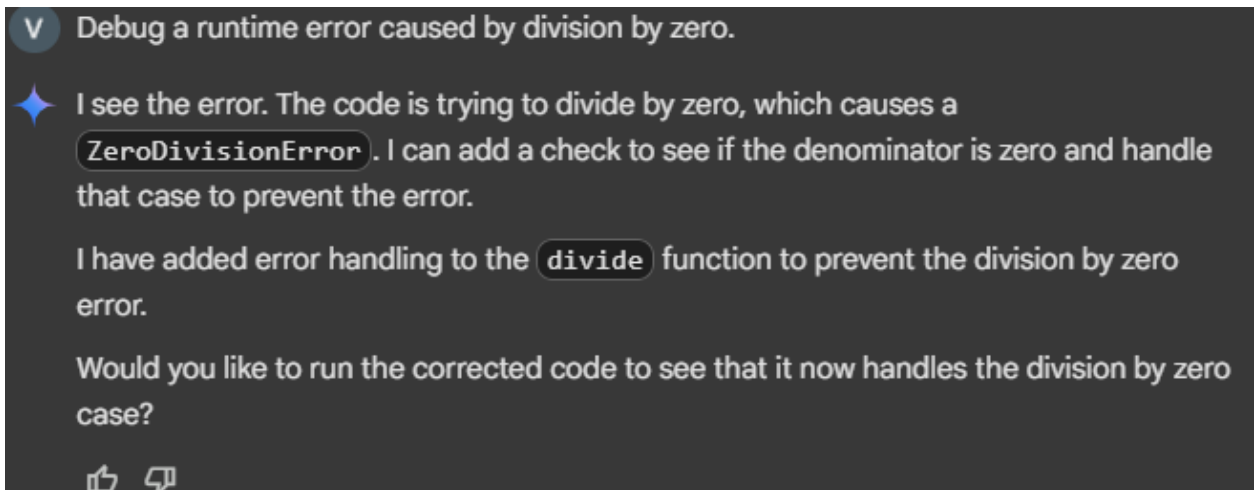
- Debug a runtime error caused by division by zero. Let AI insert try-except.

CODE:



```
def divide(a,b):  
    return a/b  
  
print(divide(10,0))
```

PROMPT:



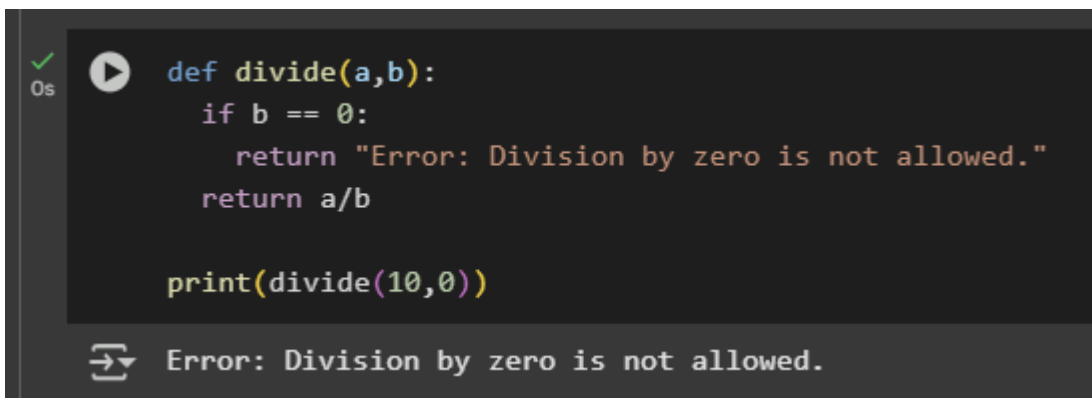
V Debug a runtime error caused by division by zero.

✨ I see the error. The code is trying to divide by zero, which causes a `ZeroDivisionError`. I can add a check to see if the denominator is zero and handle that case to prevent the error.

I have added error handling to the `divide` function to prevent the division by zero error.

Would you like to run the corrected code to see that it now handles the division by zero case?

OUTPUT:



```
def divide(a,b):  
    if b == 0:  
        return "Error: Division by zero is not allowed."  
    return a/b  
  
print(divide(10,0))
```

Error: Division by zero is not allowed.

Task Description#4

Provide a faulty class definition (missing self in parameters). Let AI fix it.

CODE:

```
✓ 0s  class Rectangle:
    def __init__(length, width):
        self.length = length
        self.width = width
```

PROMPT:

• Provide a faulty class definition (missing self in parameters).

✦ I have already provided a faulty class definition with the missing `self` parameter in the `__init__` method in cell `bx1l2tg2FtDS`.

Is there anything else I can help you with regarding this class or another task?

OUTPUT:

```
✓ 0s  class Rectangle:
    def __init__(self, length, width):
        self.length = length
        self.width = width

    def area(self):
        return self.length * self.width

    def perimeter(self):
        return 2 * (self.length + self.width)
```

```
✓ 0s [25] my_rectangle = Rectangle(10, 5)
      print(f"Area: {my_rectangle.area()}")
      print(f"Perimeter: {my_rectangle.perimeter()}")
```

⇒ Area: 50
Perimeter: 30

Task Description#5

- Access an invalid list index and use AI to resolve the Index Error.

CODE:

```
numbers=[1,2,3]
print(numbers[5])

-----
IndexError                                Traceback (most recent call last)
/tmp/ipython-input-3809996345.py in <cell line: 0>()
      1 numbers=[1,2,3]
----> 2 print(numbers[5])

IndexError: list index out of range
```

PROMPT:

```
V Access an invalid list index and use AI to resolve the Index Error.

✦ Okay, I will generate code that attempts to access an invalid list index, which
will cause an IndexError.
```

OUTPUT:

```
[34] my_list = [10, 20, 30]
      index_to_access = 2 # This index is out of range

      try:
          print(my_list[index_to_access])
      except IndexError:
          print(f"Error: Index {index_to_access} is out of range for the list.")
```